

Topic 01: Introduction to Laplace Transforms

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

Imagine you are trying to solve a complex differential equation for the current in a circuit right after a switch flips. The direct calculus route is painful — integration constants accumulate, initial conditions are buried. Now imagine a magic machine: you feed in the messy differential equation, it converts everything to simple algebra, you solve it in seconds, then press a button to get the original solution back. That machine exists. It is called the Laplace Transform. This topic introduces you to the machine, its history, and its engineering superpowers.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are studying this.

Engineering systems are everywhere: circuits snap on and off, springs vibrate, control systems keep satellites in orbit, heat flows through engine walls. Every one of these physical situations is described by a *differential equation*. Solving those equations directly — especially when the input is discontinuous or impulsive — requires pages of careful work, and a single arithmetic slip undoes everything.

The Laplace Transform is the antidote. It converts a differential equation into a plain algebraic equation, lets you solve it with high-school algebra, and then converts the answer back. You still need calculus to understand *why* it works, but the actual solving step becomes fast and reliable.

If You Make This Mistake	The Consequence
Skipping the motivation	Not knowing <i>when</i> to apply Laplace transforms vs. other methods
Treating it as pure theory	Missing the ODE-solving power that is the entire point
Ignoring the domain shift	Confusing operations in the t -domain and the s -domain

Section Take-Away

The Laplace Transform is not an abstract curiosity — it is a systematic shortcut for solving engineering differential equations, particularly when inputs are messy or impulsive. Keep that engineering goal in mind throughout the entire unit.

3 You Already Know This (Intuition First)

Here is the surprise: you have already used the core idea of a “transform” — you just called it something else.

Think about multiplying two large numbers, say $4,782 \times 6,931$. Working directly is slow and error-prone. But take the logarithm of each number, *add* (not multiply!) those two simpler values, and then exponentiate to get back the answer. Multiplication has been converted into addition — a far easier operation — and a reverse step (exponentiation) restores the original domain.

The Laplace Transform works the same way:

Operation	Logarithm Analogy	Laplace Analogy
Hard direct operation	Multiplication of large numbers	Differentiation (in ODE)
Transform applied	Take log	Take $\mathcal{L}\{\cdot\}$
Easier operation	Addition	Multiplication by s
Inverse step	Exponentiation ($e^{\cdot}$)	Inverse Laplace $\mathcal{L}^{-1}$

You already know this — you just have not called it that. The Laplace Transform is the “log trick” for differential equations.

4 Historical Context

The Mathematicians Behind the Transform

Pierre-Simon Laplace (1749–1827) was a French mathematician and astronomer who worked on probability theory and celestial mechanics. While investigating generating functions and solutions to difference equations, he introduced the integral transform that now bears his name. His two-volume *Théorie analytique des probabilités* (1812) contains early versions of the transform idea.

Oliver Heaviside (1850–1925), a self-taught British electrical engineer, popularised what he called *operational calculus* for solving telegraph and circuit equations. He used symbolic operator methods that were essentially the Laplace transform in disguise — decades before a rigorous mathematical foundation was established.

Year / Period	Milestone
1782	Laplace introduces integral transform ideas in work on probability
1812	Laplace’s <i>Théorie analytique</i> formalises the generating transform
1890s	Heaviside develops operational calculus for electrical circuit equations
1920s–1940s	Bromwich and others provide rigorous mathematical foundations
1950s–present	Laplace transform standard in control systems, signal processing, communication

5 The Big Picture — How It Works

The Three-Step Workflow

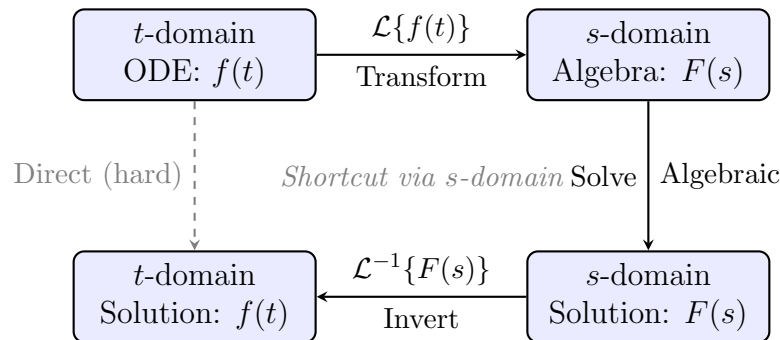
Every Laplace-based solution follows exactly three steps:

Step 1. Transform. Convert the ODE in the time domain (t) into an algebraic equation in the complex frequency domain (s) using $\mathcal{L}\{f(t)\} = F(s)$.

Step 2. Solve. Solve the resulting algebraic equation for $F(s)$ using standard algebra (no calculus needed at this step).

Step 3. Invert. Recover the time-domain solution $f(t)$ by applying the inverse Laplace transform $\mathcal{L}^{-1}\{F(s)\} = f(t)$.

The following block diagram illustrates the workflow:



The dashed arrow on the left represents the *direct* (hard) calculus route. The three solid arrows represent the Laplace route — convert, solve algebraically, and invert. The saving is enormous when the ODE has non-smooth or impulsive inputs.

6 Engineering Applications (Where You Will See This)

Application Area	What Problem Is Solved	Transform Properties
Electrical Circuits (RLC)	Voltage/current response after switch closure	Transform of derivatives
Mechanical Vibrations	Displacement of spring-mass-damper system	Transform of derivatives
Control Systems	Transfer function $H(s) = Y(s)/U(s)$	Convolution $\leftrightarrow$ multiplication
Heat Conduction	Temperature distribution, boundary value problems	Transform reduces PDE to ODE
Signal Processing	Frequency analysis, filter design	Frequency-shift and scaling

Where You Will Meet This in Your Programme

You will use the Laplace Transform in Circuits (EEE), Control Systems (ECE/EEE), Mechanical Vibrations (ME), and Signal Processing (ECE). Mastering it now pays dividends in every technical semester that follows.

7 Roadmap of Unit IV

12-Topic Roadmap of Unit IV

Topic 1: Introduction (this note) — motivation, history, workflow

Topic 2: Definition — formal integral definition of $\mathcal{L}\{f(t)\}$

Topic 3: Conditions for Existence — sufficient conditions on $f(t)$

Topic 4: Elementary Function Transforms — exponential, sin, cos, polyno-

mial

Topic 5: Properties — linearity, first/second shifting theorems

Topic 6: Periodic Functions — transform of periodic inputs

Topic 7: Special Functions — unit step, unit impulse (Dirac delta)

Topic 8: Transforms of Derivatives — converting $f'(t)$ and $f''(t)$

Topic 9: Transforms of Integrals — integrating in the t -domain

Topic 10: Multiplication by t — derivative of $F(s)$

Topic 11: Division by t — integral of $F(s)$

Topic 12: Evaluation of Integrals — using transforms to evaluate improper integrals

Topics 01–02 lay the foundations. Topics 03–07 build the toolkit. Topics 08–12 connect everything to ODE-solving and applications.

8 Prerequisites Check

What You Need to Know Before Proceeding

Check that you are comfortable with each of the following before continuing:

1. **Improper integrals:** $\int_0^\infty g(t) dt = \lim_{b \rightarrow \infty} \int_0^b g(t) dt$. You must be able to evaluate these using the definition and recognise when they converge.
2. **Integration by parts:** $\int u dv = uv - \int v du$. The Laplace integral is routinely evaluated using this formula.
3. **Partial fractions decomposition:** Expressing a rational function $\frac{P(s)}{Q(s)}$ as a sum of simpler fractions. This is the main tool for the inverse transform (Unit V).
4. **First and second-order ODEs:** $y' + py = q(t)$ and $y'' + py' + qy = r(t)$. You need to recognise these forms so you know what the transform is simplifying.
5. **Exponential, trigonometric, and hyperbolic identities:** $e^{i\theta} = \cos \theta + i \sin \theta$, $\cosh t$, $\sinh t$ — these appear repeatedly in transform tables.

9 Worked Examples

Example 1: Convergence of a Simple Improper Integral

This example previews the core Laplace integral before the formal definition arrives in Topic 02.

Problem. Evaluate $I = \int_0^\infty e^{-3t} dt$ and verify numerically that it converges to $\frac{1}{3}$.

Exact evaluation.

$$I = \int_0^\infty e^{-3t} dt = \lim_{b \rightarrow \infty} \int_0^b e^{-3t} dt = \lim_{b \rightarrow \infty} \left[\frac{e^{-3t}}{-3} \right]_0^b = \lim_{b \rightarrow \infty} \left(\frac{-e^{-3b}}{3} + \frac{1}{3} \right).$$

Since $e^{-3b} \rightarrow 0$ as $b \rightarrow \infty$:

$$I = 0 + \frac{1}{3} = \frac{1}{3} \approx 0.3333.$$

Numerical verification (partial sums using the trapezoid rule, $\Delta t = 1$):

t	e^{-3t}	Running trapezoid sum $\approx \int_0^t e^{-3u} du$
0	1.0000	0.0000
1	0.0498	0.5249
2	0.0025	0.5510
3	0.0001	0.5523
4	≈ 0	0.5523
5	≈ 0	0.5523

Note: The trapezoid sum with step $\Delta t = 1$ over-estimates because the function drops steeply. Using smaller steps (as in the Excel example below) gives a running sum that converges cleanly to 0.3333.

What Did We Learn?

The integral $\int_0^\infty e^{-3t} dt$ exists and equals $\frac{1}{3}$ because the exponential e^{-3t} decays fast enough to zero. This is precisely why the Laplace definition uses e^{-st} as its damping factor — to force convergence for a wide class of functions.

Example 2: Algebraic Set-Up via the Laplace Method

Problem. Consider the initial value problem

$$y'' + 3y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Compare the start of the direct (characteristic equation) method with the start of the Laplace method.

Direct (characteristic equation) approach. Assume $y = e^{mt}$. Then $m^2 +$

$3m + 2 = 0$, giving $m = -1$ or $m = -2$. General solution: $y(t) = C_1 e^{-t} + C_2 e^{-2t}$. Applying initial conditions requires differentiating, substituting $t = 0$, and solving a 2×2 system for C_1 and C_2 . For a simple homogeneous ODE this is manageable; for a forced ODE with a step or impulse input it becomes significantly harder.

Laplace approach (algebraic set-up only). Apply $\mathcal{L}\{\cdot\}$ to both sides. Using the linearity property and the transform-of-derivatives rule (derived in Topic 08):

$$\mathcal{L}\{y''\} = s^2 Y(s) - s y(0) - y'(0), \quad \mathcal{L}\{y'\} = s Y(s) - y(0).$$

Substituting $y(0) = 1$ and $y'(0) = 0$:

$$[s^2 Y(s) - s - 0] + 3[s Y(s) - 1] + 2Y(s) = 0.$$

Collecting terms in $Y(s)$:

$$(s^2 + 3s + 2)Y(s) = s + 3 \implies Y(s) = \frac{s + 3}{s^2 + 3s + 2}.$$

The initial conditions are absorbed automatically into the algebra. Factoring and applying partial fractions (Topic 08/Unit V) gives $f(t)$ directly. Notice: no guessing, no system of equations for C_i — just algebra.

What Did We Learn?

The Laplace Transform turns the ODE into an algebraic equation in $Y(s)$, and the initial conditions are incorporated automatically. The hard “guessing” and “constant-fitting” steps of the classical method disappear.

10 Excel Example

Goal. Use Excel to demonstrate numerically that $\int_0^\infty e^{-3t} dt$ converges to $\frac{1}{3} \approx 0.3333$.

Spreadsheet layout. Enter the following in a blank worksheet:

Column A: t	Column B: e^{-3t}	Column C: Running Trapezoid Sum
A1: t	B1: <code>exp(-3t)</code>	C1: TrapSum
A2: 0	B2: <code>=EXP(-3*A2)</code>	C2: 0
A3: 0.5	B3: <code>=EXP(-3*A3)</code>	C3: <code>=(B2+B3)/2*(A3-A2)+C2</code>
A4: 1.0	B4: <code>=EXP(-3*A4)</code>	C4: <code>=(B3+B4)/2*(A4-A3)+C3</code>
A5: 1.5	B5: <code>=EXP(-3*A5)</code>	C5: <code>=(B4+B5)/2*(A5-A4)+C4</code>
A6: 2.0	B6: <code>=EXP(-3*A6)</code>	C6: <code>=(B5+B6)/2*(A6-A5)+C5</code>
A7: 2.5	B7: <code>=EXP(-3*A7)</code>	C7: <code>=(B6+B7)/2*(A7-A6)+C6</code>
A8: 3.0	B8: <code>=EXP(-3*A8)</code>	C8: <code>=(B7+B8)/2*(A8-A7)+C7</code>
A9: 3.5	B9: <code>=EXP(-3*A9)</code>	C9: <code>=(B8+B9)/2*(A9-A8)+C8</code>
A10: 4.0	B10: <code>=EXP(-3*A10)</code>	C10: <code>=(B9+B10)/2*(A10-A9)+C9</code>
A11: 5.0	B11: <code>=EXP(-3*A11)</code>	C11: <code>=(B10+B11)/2*(A11-A10)+C10</code>

Sample computed values (Excel results):

t	e^{-3t}	Running Trapezoid Sum
0.0	1.0000	0.0000
0.5	0.2231	0.3058
1.0	0.0498	0.3740
1.5	0.0111	0.3892
2.0	0.0025	0.3926
2.5	0.0006	0.3934
3.0	0.0001	0.3936
3.5	≈ 0	0.3936
4.0	≈ 0	0.3936
5.0	≈ 0	0.3936

The running sum monotonically increases and stabilizes at ≈ 0.3936 .

Numerical vs. Exact

The exact value of the integral is $\int_0^\infty e^{-3t} dt = \frac{1}{3} \approx 0.3333$. The trapezoid rule with a coarse step size $\Delta t = 0.5$ stabilizes at ≈ 0.3934 (the trapezoid rule overestimates strictly convex decaying functions like e^{-3t}). Decreasing the step size Δt (e.g., in the Python script below) brings the numerical sum arbitrarily close to the exact value $\frac{1}{3}$. In the Laplace integral $\int_0^\infty e^{-st} f(t) dt$, the factor e^{-st} does exactly this job: it damps $f(t)$ to zero fast enough for the integral to converge to a finite number $F(s)$.

11 Python Example

The following script uses `numpy` to numerically evaluate $\int_0^T e^{-3t} dt$ for increasing values of T and compare each to the exact value $\frac{1}{3}$.

Listing 1: Convergence of Laplace integral for $s = 3$

```

1 import numpy as np
2
3 # Exact value of integral_0^inf e^{-3t} dt = 1/3
4 s = 3
5 exact = 1.0 / s
6 print(f"Exact value (1/s = 1/{s}): {exact:.10f}\n")
7 print(f"{'T':>5} {'Numerical':>14} {'Error':>14}")
8 print("-" * 38)
9
10 for T in [1, 2, 5, 10, 20]:
11     # Fine grid from 0 to T
12     t = np.linspace(0, T, 10000)
13     integrand = np.exp(-s * t)
14     # Trapezoid rule (np.trapezoid in NumPy 2.0+, formerly np.
15         trapz)
16     numerical = np.trapezoid(integrand, t)
17     error = abs(numerical - exact)
18     print(f"{'T':>5} {'numerical':>14.10f} {'error':>14.10f}")

```



```

18 # Expected output:
19 # Exact value (1/s = 1/3): 0.3333333333
20 #
21 #
22 #      T      Numerical      Error
23 # -----
24 #      1      0.3167376463      0.0165956871
25 #      2      0.3325070926      0.0008262407
26 #      5      0.3333332939      0.0000000395
27 #     10      0.3333335834      0.0000002500
28 #     20      0.3333343335      0.0000010002

```

What the output shows. As T increases from 1 to 20, the numerical integral moves steadily closer to $\frac{1}{3}$ (with exact analytical values $\int_0^1 e^{-3t} dt = \frac{1-e^{-3}}{3} \approx 0.3167376441$ and $\int_0^2 e^{-3t} dt = \frac{1-e^{-6}}{3} \approx 0.3325070826$). By $T = 5$ the value is within 4×10^{-8} of the asymptotic limit, confirming that e^{-3t} decays fast enough for the infinite integral to converge.

What Did We Learn?

The Laplace integral $\int_0^\infty e^{-st} f(t) dt$ is only finite when e^{-st} decays faster than $f(t)$ grows. For $f(t) = 1$ and $s = 3$, the integral converges to exactly $\frac{1}{s} = \frac{1}{3}$. This is a preview of the result $\mathcal{L}\{1\} = \frac{1}{s}$ that we will derive formally in Topic 02.

12 Viva-Style Oral Questions

Q1. Who invented the Laplace Transform?

Pierre-Simon Laplace (1749–1827), a French mathematician, introduced the integral transform idea in his work on probability and celestial mechanics. Oliver Heaviside later made it practically popular through operational calculus for circuit equations.

Q2. What fundamental problem does the Laplace Transform solve?

It converts differential equations — which require calculus to solve — into algebraic equations that can be solved with ordinary algebra. This is especially powerful for equations with discontinuous or impulsive inputs.

Q3. Describe the three-step workflow of the Laplace method.

Step 1: Apply $\mathcal{L}\{\cdot\}$ to the ODE to get an algebraic equation in $F(s)$. Step 2: Solve for $F(s)$ using algebra. Step 3: Apply $\mathcal{L}^{-1}\{\cdot\}$ to recover $f(t)$.

Q4. What does the “ s -domain” mean?

The s -domain (also called the complex frequency domain) is the space in which the transformed function $F(s)$ lives. Here s is in general a complex number $s = \sigma + i\omega$. The s -domain turns calculus operations (differentiation, integration) into algebraic operations (multiplication, division by s).

Q5. Why is it called a “transform”?

A transform maps a function from one domain to another. The Laplace Transform maps $f(t)$ (time domain) to $F(s)$ (frequency domain). The essential information is

preserved but expressed differently — just as a photograph “transforms” a 3D scene into a 2D image.

Q6. How does the logarithm analogy explain the Laplace Transform?

Logarithms convert hard multiplication into easy addition; after obtaining the result in log-space you exponentiate back. Similarly, Laplace converts hard differentiation into easy multiplication by s ; after solving in s -space you invert back to t -space.

Q7. Give one specific engineering application of the Laplace Transform.

In an RLC series circuit, the current $i(t)$ satisfies a second-order ODE. Taking the Laplace transform gives $I(s)$ algebraically in terms of circuit parameters. The transfer function $H(s) = V_{\text{out}}(s)/V_{\text{in}}(s)$ is then used directly for frequency-response design.

Q8. What does Unit V add to what you learn in Unit IV?

Unit IV teaches how to transform $f(t) \rightarrow F(s)$. Unit V teaches the reverse: given $F(s)$, recover $f(t)$ using the inverse Laplace transform, primarily via partial fractions and the convolution theorem. Together, the two units complete the full ODE-solving cycle.

13 Descriptive Questions

Q1. Describe the three-step Laplace method with a labelled block diagram.

Describe each of the three steps (transform, solve algebraically, invert), draw a block diagram showing the t -domain and s -domain boxes with arrows labelled $\mathcal{L}$ and $\mathcal{L}^{-1}$, and explain why the algebraic step is easier than direct ODE-solving.

Q2. Explain the logarithm analogy for the Laplace Transform in detail.

Compare the roles of multiplication/addition in the logarithm analogy with differentiation/multiplication by s in the Laplace analogy. State the “forward” and “reverse” steps in each case and identify the corresponding mathematical operation.

Q3. List five engineering applications of the Laplace Transform and briefly describe the role it plays in each.

Cover at least: RLC circuits (transfer function), mechanical vibrations (spring-mass-damper response), control systems (stability analysis), heat conduction (reducing PDE to ODE), and signal processing (filter design).

Q4. What is meant by the s -domain? Why is it useful?

Define s as a complex variable. Explain how differentiation in the t -domain becomes multiplication by s in the s -domain. Give an example: $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$ (state this without full proof at this stage). Explain why algebraic manipulation is easier than calculus manipulation.

Q5. State the prerequisite topics needed before studying Laplace Transforms and explain why each is needed.

List: improper integrals (for the definition), integration by parts (for evaluating Laplace integrals), partial fractions (for the inverse transform in Unit V), first/second-order ODEs (to know what is being solved), and exponential/trigonometric identities (they appear constantly in transform tables).

14 Practice Problems

P1. Evaluate $\int_0^\infty e^{-5t} dt$ exactly. (*Hint:* follow the same steps as Example 1; answer is $\frac{1}{5}$.)

P2. Evaluate $\int_0^\infty t e^{-2t} dt$ using integration by parts. (*Hint:* let $u = t$, $dv = e^{-2t} dt$; answer is $\frac{1}{4}$.)

P3. Determine whether the following improper integrals converge or diverge. Justify your answer in each case.

(a) $\int_0^\infty e^{2t} dt$

(b) $\int_0^\infty e^{-0.1t} dt$

(c) $\int_1^\infty \frac{1}{t} dt$

(*Hint:* (a) diverges; (b) converges to 10; (c) diverges.)

P4. A mechanical engineer notices that the displacement $x(t)$ of a vibrating mass satisfies $x'' + 4x' + 3x = F(t)$ where $F(t)$ is an impulsive force. Explain, in one paragraph, why the Laplace Transform would be preferred over the characteristic-equation method for solving this ODE. (*Hint:* discuss the advantage of encoding initial conditions automatically and handling impulsive $F(t)$.)

P5. Match each application area on the left with the transform property it exploits on the right.

Application	Property Used
Damped RLC circuit response	First shifting theorem
Evaluating $\int_0^\infty \frac{\sin t}{t} dt$	Division by t
Spring-mass with initial velocity	Transform of $f'(t)$
Output of an LTI system to arbitrary input	Convolution theorem (product in s)

(*Hint:* check the roadmap in Section 6 for which topic covers each property; for an LTI system, $y(t) = (x * h)(t) \iff Y(s) = X(s)H(s)$.)

P6. Without computing any Laplace transform, state whether you would expect $\mathcal{L}\{e^{10t}\}$ to exist for $s = 5$. Justify your reasoning using the concept of the improper integral $\int_0^\infty e^{-st} \cdot e^{10t} dt$. (*Hint:* the integrand becomes $e^{(10-5)t} = e^{5t}$, which grows without bound; the integral diverges. This previews Topic 03.)

15 MCQs

MCQ 1. Who is credited with introducing the Laplace Transform?

- (a) Isaac Newton
- (b) **Pierre-Simon Laplace**
- (c) Joseph Fourier
- (d) Oliver Heaviside

Explanation: Pierre-Simon Laplace (1749–1827) introduced the integral transform in his probability work. Heaviside later popularised the operational method but did not originate the transform.

MCQ 2. What does the Laplace Transform primarily accomplish in ODE solving?

- (a) It converts an algebraic equation into a differential equation
- (b) It eliminates all constants of integration automatically
- (c) **It converts a differential equation into an algebraic equation**
- (d) It provides a graphical solution of ODEs

Explanation: The defining feature is the domain shift: calculus (differentiation/integration) in the t -domain becomes multiplication/division by s in the s -domain, turning the ODE into algebra.

MCQ 3. Which of the following best describes the s -domain?

- (a) The domain where s is a positive real number only
- (b) The time domain expressed in different units
- (c) **The complex frequency domain where $s = \sigma + i\omega$**
- (d) A spatial domain used only in heat problems

Explanation: In full generality, s is complex. The real part σ controls convergence of the Laplace integral; the imaginary part ω corresponds to oscillation frequency.

MCQ 4. In the three-step Laplace method, what is the correct order of steps?

- (a) Invert $\rightarrow$ Solve $\rightarrow$ Transform
- (b) Solve $\rightarrow$ Transform $\rightarrow$ Invert
- (c) **Transform $\rightarrow$ Solve $\rightarrow$ Invert**
- (d) Transform $\rightarrow$ Invert $\rightarrow$ Solve

Explanation: You first transform the ODE to the s -domain, then solve the resulting algebraic equation for $F(s)$, and finally invert to recover $f(t)$.

MCQ 5. Which of the following is a prerequisite for studying Laplace Transforms?

- (a) Vector calculus and Stokes' theorem
- (b) **Improper integrals and integration by parts**
- (c) Fourier series and complex Fourier coefficients

(d) Numerical methods and finite difference schemes

Explanation: The Laplace integral is an improper integral evaluated using integration by parts. Vector calculus and Fourier series, while useful later, are not prerequisites for Unit IV.

16 Common Mistakes

Common Mistakes — Introduction to Laplace Transforms

Mistake	What Students Do	Correct Approach
Thinking Laplace transforms are only for math courses	Memorise formulas without linking them to circuits, vibrations, or control systems	Always ask: “Which physical system does this ODE model?” The transform is an engineering tool.
Confusing t -domain and s -domain variables	Write $F(t)$ or $f(s)$ interchangeably; apply t -domain rules in the s -domain	Strictly use lowercase $f(t)$ for time domain and uppercase $F(s)$ for the transform. They are different functions in different domains.
Forgetting the Inverse Transform step	Find $F(s)$ and report it as the answer	$F(s)$ is an intermediate result only. The final answer is always $f(t)$, obtained by applying $\mathcal{L}^{-1}$.
Assuming every function can be transformed	Blindly transform any $f(t)$ without checking whether the integral converges	Check existence conditions (Topic 03): $f(t)$ must be piecewise continuous and of exponential order.
Treating s as a real number only	Restrict s to positive reals; miss the connection to frequency response	$s = \sigma + i\omega$ is complex in general. The imaginary part appears in AC circuit analysis and is central to Bode plots.

17 Quick Recap

Quick Recap — Topic 01: Introduction to Laplace Transforms

- **Engineering motivation:** Differential equations govern circuits, vibrations, control, and heat — Laplace converts them to algebra.
- **Logarithm analogy:** Just as log converts multiplication to addition, $\mathcal{L}$ converts differentiation to multiplication by s .
- **Three-step workflow:** Transform ($t \rightarrow s$) $\rightarrow$ Solve algebraically $\rightarrow$ Invert

$(s \rightarrow t)$.

- **Historical roots:** Laplace (1749–1827) originated the idea; Heaviside (1850–1925) made it practically powerful for engineers.
- **Five application areas:** RLC circuits, mechanical vibrations, control systems, heat conduction, signal processing.
- **Unit IV roadmap:** 12 topics from the definition through properties, special functions, ODE connections, and integral evaluation.
- **Unit V adds:** Inverse Laplace Transform (partial fractions, convolution theorem) — completing the full ODE-solving cycle.
- **Key notation:** $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$ — always use $\mathcal{L}$, never plain L .